LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	22
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Stability and control: static, dynamic stability
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Stability and control: static, dynamic stability. definition, notation used Stability requirements, a/c controls and Manoeuvres,
5. Specific Objectives	11.1.4.1 Trainee will be able to describe aircraft stability in all the three axis

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of static stability definitions and dynamic stability concepts using IB and L & Ds; Instructor supervision required for concept verification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Define the term 'stability' in the context of aircraft motion.
(b) List the three axes of aircraft rotation.
(c) State the primary difference between static and dynamic stability.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of static stability • Notation used for stability axes • Static stability requirements • Aircraft controls and manoeuvres	L & Ds	PPT, IB	15 min
(b)	• Definition of dynamic stability • Oscillatory motion characteristics • Damping effects on stability • Comparison of static and dynamic responses	L & Ds	PPT, PDF	20 min
(c)	• Summary of stability in three axes • Key stability requirements recap • Role of controls in stability • Link to next lesson on longitudinal stability	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how dynamic stability differs from static stability regarding oscillations.
(b) Identify the notation used for the three axes of stability.
(c) Describe the relationship between aircraft controls and stability requirements.

12. Home Assignment and Follow Up	Prepare a brief note on the forces and moments affecting static longitudinal stability as covered in LP 23.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Static longitudinal stability: forces and moments Stick fixed and stick free stability, neutral point and static margin, stability curve, Longitudinal dynamic stability,	Refer to the master syllabus reference list	LP 23

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min